

number		
3(a)(i)	A α glucose molecules join together by a condensation reaction B is incorrect because bonds form by condensation reactions C is incorrect because α glucose molecules join together D is incorrect because α glucose molecules join together by condensation reactions	(1)

Question number	Answer	Additional guidance	Mark
3(a)(ii)	glycosidic (bond / link)	Accept covalent (bond) Ignore any numbers eg 1,4 glycosidic	(1)

Question number	Answer	Additional guidance	Mark
3(a)(iii)	An explanation that includes the following points: • {polymer of glucose / polysaccharide} therefore has a high energy content (1) • {large molecule / polymer / polysaccharide} therefore {insoluble / has no osmotic effect} (1) • branched structure therefore {broken down / energy released / hydrolysis is} faster (1) • compact therefore has a high energy density (1)	Accept {large molecule / polymer} so does not diffuse Accept branched so can be broken down from several points at the same time Ignore easier to break down Accept compact so {high energy stored in a small space / many glucose molecules stored in a small space}	(3)

Question number	Answer	Additional guidance	Mark
3(b)(i)	Answer that includes the following points: - estimate of number of babies with GSD (1) - estimate of mean number of babies with Von Gierke disease (1)	Accept any value between 150 and 200 Accept any value between 38 and 50 Accept non-whole numbers eg 38.5 Correct answer with no working (2)	(2)

Question number	Answer	Additional guidance	Mark
3(b)(ii)	An answer that includes the following points: - because allele for Von Gierke disease may be recessive (1) - therefore (both) parents may be heterozygous (1) - because individuals with Von Gierke disease are less likely to have babies (1)	Accept disease caused by two recessive alleles Do not accept {gene / disease} is recessive Accept parents may be carriers (of the allele) Ignore mutation	(2)